

RED & WHITE SKILL EDUCATION

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Project: Expectation Decider

Theory Completion Report — Probability & Statistics Analysis

This report presents the theoretical explanations, formulas, and step-by-step calculations for the Expectation Decider project, analyzing probability patterns from a dataset of 200 students to determine the likelihood of passing a competitive mathematics exam.

1. Understanding the Basics

What is Probability?

Probability is a numerical measure, between 0 and 1, of how likely an event is to occur. A probability of 0 means the event is impossible, and a probability of 1 means the event is certain to happen. It is calculated as:

$$P(E) = (\text{Number of favourable outcomes}) / (\text{Total number of possible outcomes})$$

In this project, probability lets us quantify questions such as: "How likely is a student to pass the exam, given certain study habits or attendance patterns?"

Key Probability Terminology

Term	Meaning
Experiment / Trial	A process that produces an observable outcome.
Sample Space (S)	The set of all possible outcomes of an experiment.
Event (E)	A subset of the sample space we are interested in.
Random Variable	A variable whose value is a numerical outcome of a random process.
Independent Events	Occurrence of one event does not affect the probability of the other.
Dependent Events	Occurrence of one event changes the probability of the other.
Mutually Exclusive	Two events that cannot occur at the same time.
Conditional Probability	Probability of event A given event B has occurred: $P(A B)$.
Joint Probability	Probability that two events occur together: $P(A \text{ and } B)$.
Marginal Probability	Probability of a single event, ignoring all other variables.

Three Probability Event Examples from the Dataset

- Event A: A randomly selected student has attendance greater than 80%.
- Event B: A randomly selected student participated in group discussion (Yes).
- Event C: A randomly selected student's final exam result is "Pass".

Calculated basic probabilities (from all 200 students):

Event	Count	Probability
Attendance > 80%	89 / 200	0.4450
Group Discussion = Yes	125 / 200	0.6250
Final Exam Pass	150 / 200	0.7500

2. Types of Events — Empirical vs Theoretical Probability

Empirical (Experimental) Probability is derived from actually observed data / real-world frequency.

Theoretical (Classical) Probability is derived from reasoning about equally likely outcomes in an idealized sample space, without needing real data.

Empirical Probability Scenario

Event: A randomly selected student from the dataset passed the exam.

$$P(\text{Pass})_{\text{empirical}} = (\text{Number who passed}) / (\text{Total students}) = 150 / 200 = 0.75$$

Theoretical Probability Scenario

Event: Tossing a fair coin to decide which student's record to review first — probability of Heads.

$$P(\text{Heads})_{\text{theoretical}} = 1 / 2 = 0.5$$

This is theoretical because it comes from the logical structure of a fair coin (2 equally likely outcomes), not from observed data.

3. Random Variable & Probability Distribution

Let X = "Number of students who pass the final exam out of 3 randomly selected students." Since each student independently passes with probability $p = P(\text{Pass}) = 0.75$ and fails with probability $q = 1 - p = 0.25$, X follows a Binomial Distribution:

$$X \sim \text{Binomial}(n = 3, p = 0.75)$$

$$P(X = k) = C(n, k) \cdot p^k \cdot (1-p)^{(n-k)}, \text{ for } k = 0, 1, 2, 3$$

$X = k$	Calculation	$P(X = k)$
0	$C(3,0)(0.75)^0(0.25)^3$	0.0156
1	$C(3,1)(0.75)^1(0.25)^2$	0.1406
2	$C(3,2)(0.75)^2(0.25)^1$	0.4219
3	$C(3,3)(0.75)^3(0.25)^0$	0.4219

Check: sum of all probabilities = $0.0156 + 0.1406 + 0.4219 + 0.4219 = 1.0000$ ✓

Mean and Variance

$$\text{Mean: } E(X) = n \cdot p = 3 \times 0.75 = 2.25$$

$$\text{Variance: } \text{Var}(X) = n \cdot p \cdot (1-p) = 3 \times 0.75 \times 0.25 = 0.5625$$

$$\text{Standard Deviation: } SD(X) = \sqrt{\text{Var}(X)} = \sqrt{0.5625} \approx 0.75$$

4. Venn Diagram in Probability

Event A: Students who study more than 10 hours/week (study_hours > 10)

Event B: Students who attend more than 80% of classes (attendance > 80)

Region	Description	Count
Only A	Study > 10 hrs, attendance ≤ 80%	70
Only B	Attendance > 80%, study ≤ 10 hrs	39
$A \cap B$ (Both)	Study > 10 hrs AND attendance > 80%	50
Neither	Study ≤ 10 hrs AND attendance ≤ 80%	41
Total		200

[Venn Diagram — see notebook / recorded video for the visual two-circle diagram plotted using Matplotlib, showing the counts above in each region: only A = 70, only B = 39, $A \cap B$ = 50, neither = 41.]

5. Contingency Table & Probability Calculations

Contingency table of group_discussion (Yes/No) vs final_exam_pass (Pass/Fail):

Group Discussion	Fail	Pass	Total
No	20	55	75
Yes	30	95	125
Total	50	150	200

Joint Probability

$$P(\text{Group Discussion} = \text{Yes AND Pass}) = 95 / 200 = 0.4750$$

Marginal Probability

$$P(\text{Pass}) = 150 / 200 = 0.7500$$

Conditional Probability

$$P(\text{Pass} \mid \text{Group Discussion} = \text{Yes}) = (\text{Joint count}) / (\text{Group Discussion} = \text{Yes count}) = 95 / 125 = 0.7600$$

6. Understanding Relationships

Interpreting the Conditional Probability (plain language)

$P(\text{Pass} \mid \text{Group Discussion} = \text{Yes}) = 0.76$ tells us: "Out of all the students who did participate in group discussions, 76% of them actually passed the exam." It narrows the focus to only the sub-group satisfying the condition, and asks what proportion of that specific sub-group passed — rather than looking at the whole class.

Independent, Dependent, or Mutually Exclusive?

Two events A and B are independent only if $P(A \cap B) = P(A) \times P(B)$, which is equivalent to $P(\text{Pass} \mid \text{Group Discussion} = \text{Yes}) = P(\text{Pass})$.

$P(\text{Group Discussion} = \text{Yes}) \times P(\text{Pass}) = 0.625 \times 0.75 = 0.4688$ (expected joint probability if independent)

Actual observed $P(\text{Group Discussion} = \text{Yes AND Pass}) = 0.4750$

Comparing $P(\text{Pass} | \text{Group Discussion} = \text{Yes}) = 0.76$ with the overall $P(\text{Pass}) = 0.75$, the two values are very close (a difference of only 0.01). This suggests the events are approximately independent — participating in group discussion does not strongly change the probability of passing, at least based on this sample. The events are clearly NOT mutually exclusive, since a positive joint probability (0.475) was observed — many students both participate in group discussions and pass.

7. Bayes' Theorem Application

Given (historical data):

- $P(\text{High Attendance} | \text{Pass}) = 0.70$
- $P(\text{High Attendance} | \text{Fail}) = 0.40$
- $P(\text{High Attendance}) = 0.60$
- $P(\text{Pass})$ [prior, from dataset] = 0.75

Find: $P(\text{Pass} | \text{High Attendance})$

Bayes' Theorem:

$$P(\text{Pass} | \text{High Attendance}) = [P(\text{High Attendance} | \text{Pass}) \times P(\text{Pass})] / P(\text{High Attendance})$$

$$P(\text{Pass} | \text{High Attendance}) = (0.70 \times 0.75) / 0.60 = 0.525 / 0.60 = 0.8750$$

Interpretation: Given a student has high attendance (>80%), the probability that they passed the exam is 0.875, i.e. 87.5%. Since this is notably higher than the overall base rate of passing (75%), high attendance is a strong positive indicator of exam success.

Final Summary

Key factors most affecting the probability of passing the exam:

- **Attendance** — Students with attendance above 80% show a meaningfully higher probability of passing (87.5% via Bayes' Theorem, versus a 75% overall base rate). Attendance is one of the strongest predictors of success.
- **Group Discussion Participation** — The conditional probability $P(\text{Pass} \mid \text{Group Discussion} = \text{Yes}) = 0.76$ is close to the overall $P(\text{Pass}) = 0.75$, indicating group discussion participation and passing are approximately independent in this dataset.
- **Study Hours** — The Venn diagram shows 50 students overlap in both high study hours (>10 hrs/week) and high attendance (>80%) — this combined-effort group represents a high-likelihood-of-passing segment.
- **Previous Test Score** — Not directly modeled with a formal probability calculation above, but can be explored further (e.g. via correlation or logistic regression) as an additional predictor of `final_exam_pass`.

Overall takeaway: Passing the exam is not governed by a single factor in isolation. Attendance shows the strongest positive association with passing, while consistent study habits further compound the likelihood of success. This probability-driven analysis helps the institute identify at-risk students early (e.g., low attendance combined with low study hours) and target interventions accordingly.

Full working code, tables, and charts for every calculation above are available in `Expectation_Decider_Analysis.ipynb`, submitted alongside this report in the project repository.